Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

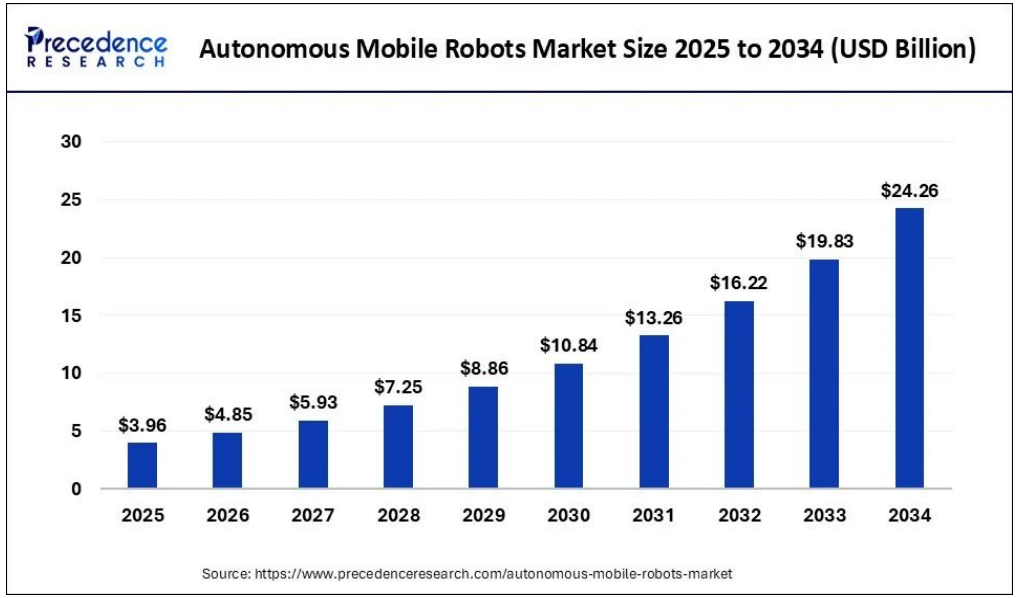
ECEN 403
Inverted Pendulum Team 3
Status Update

Team Number: 55
Team Lead: Sean Wall
Dylan Leroy, Stephen Darwin and Arian Golkarfard
TA: Swarnabha Roy
Sponsor: Aniruddha Datta

Project Summary

Problem statement:

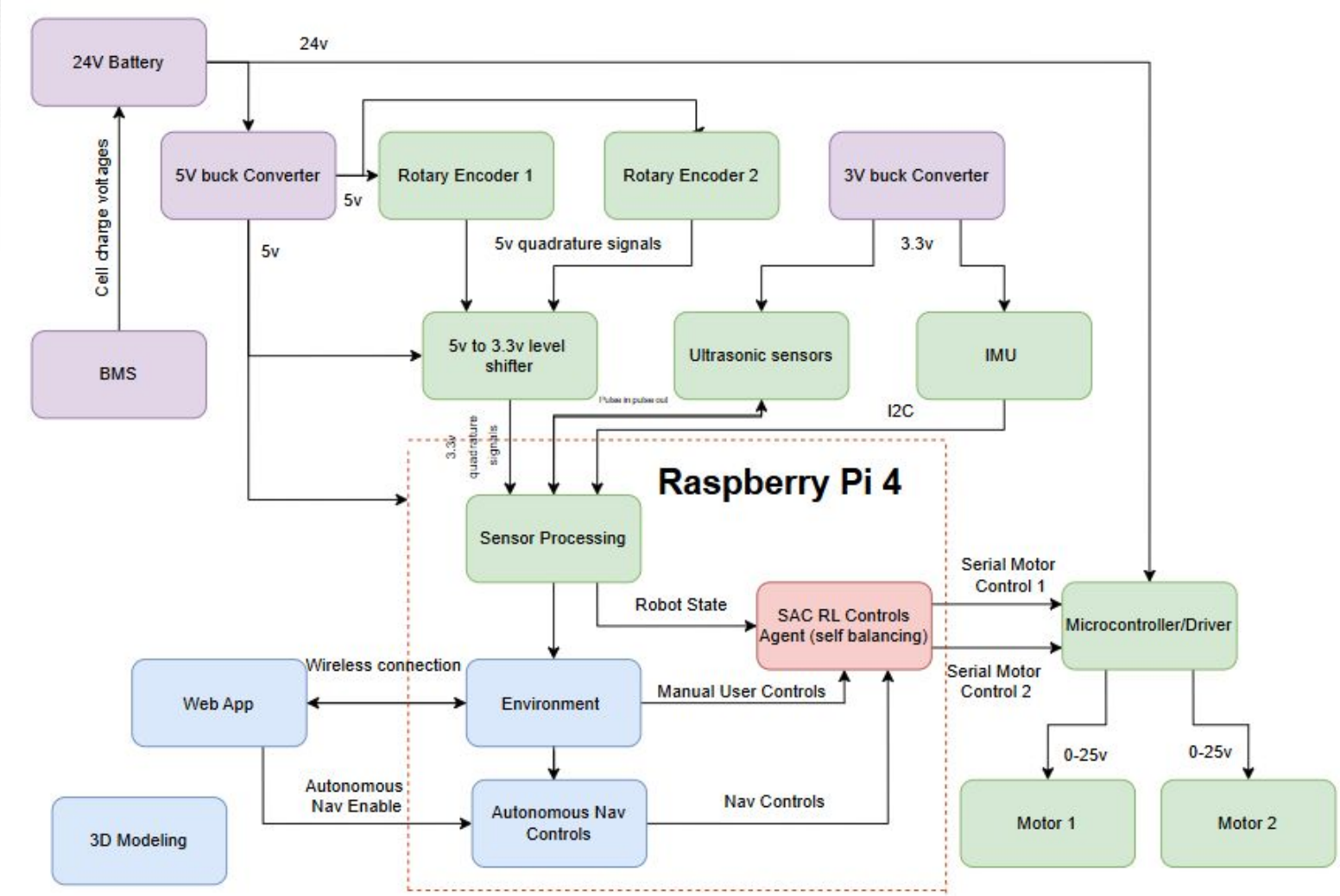
- Current mobile robot applications are rigid and inflexible
- Warehouse environments could benefit from more robust transportation methods for lightweight packages



Our Solution:

- Use reinforcement learning algorithm to train two-wheeled balancing robot
- RL will allow for a more robust and adaptable control system

System Overview



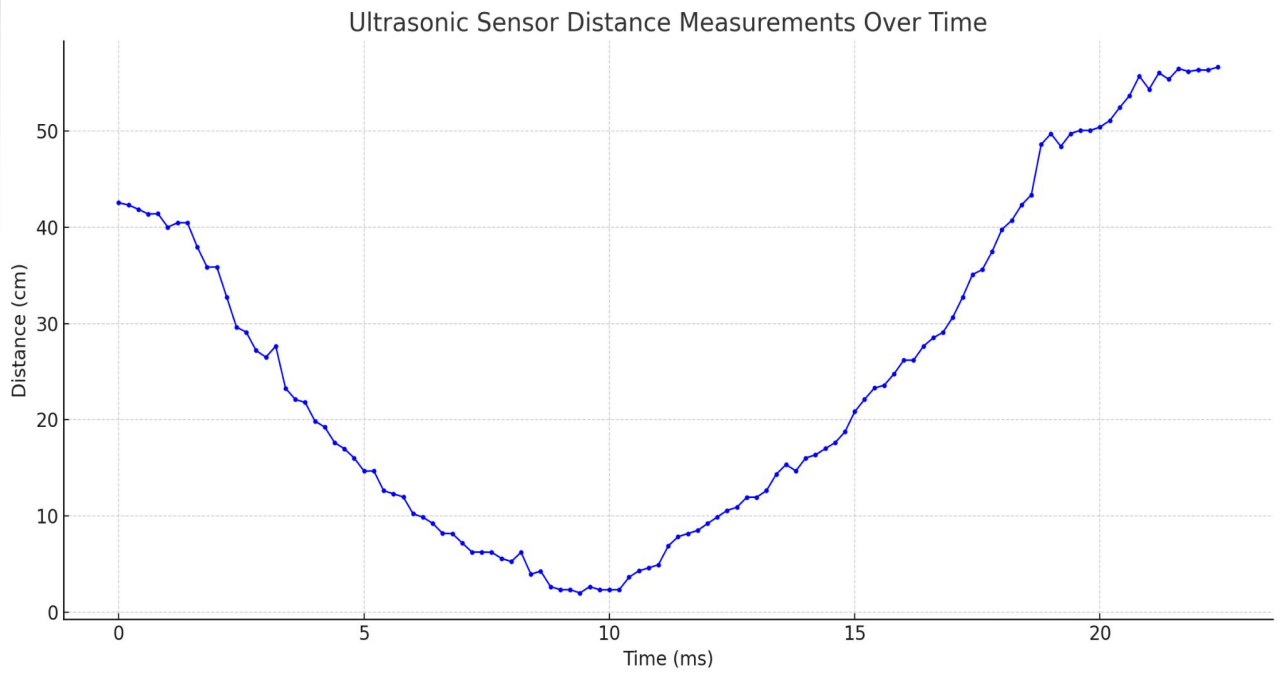
Sensor Feedback

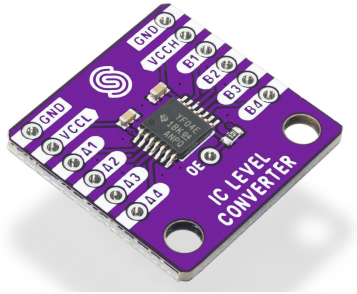
Arian Golkarfard

Accomplishments since midterm update 25-30 hours	Ongoing progress/problems and plans until the next presentation
• Verified functionality of all sensors (ultrasonic, IMU, and rotary encoders) on Raspberry Pi • Collected and analyzed plots confirming accurate and stable sensor readings • Established synchronized 200ms data refresh for integration with the SAC control agent	• Buy and test the IR sensors • Integrate all sensors to operate concurrently with synchronized data output • Lower polling rate for all sensors from 200 ms to 30 ms to improve data response • Connect sensors and power routing to new boards upon delivery • Test and calibrate timing, reduce signal noise, and optimize sensor placement

Sensor Feedback

Arian Golkarfard



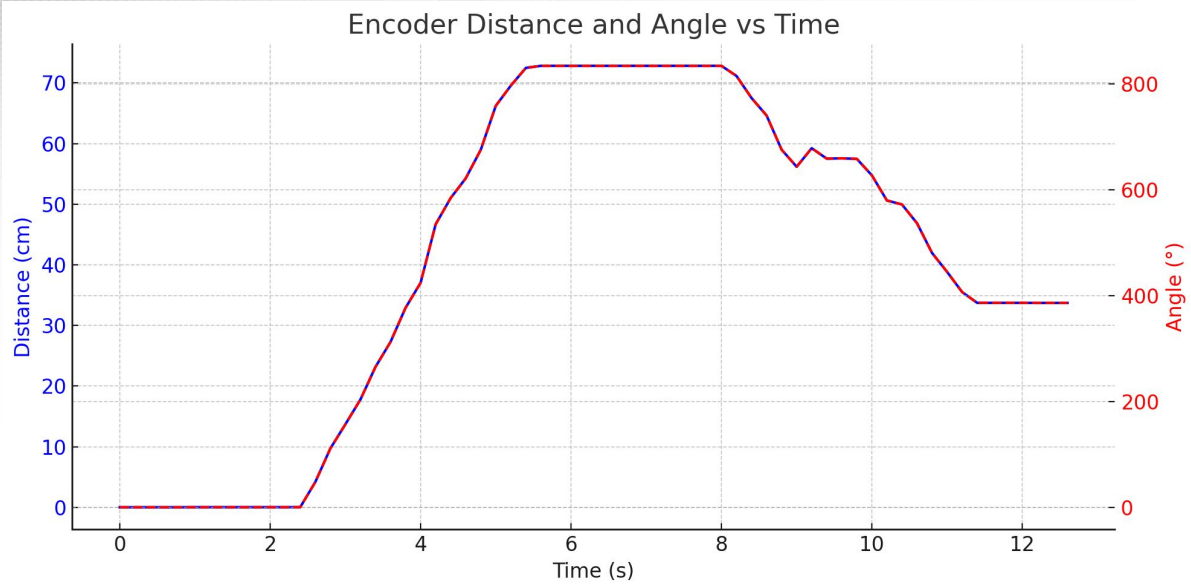


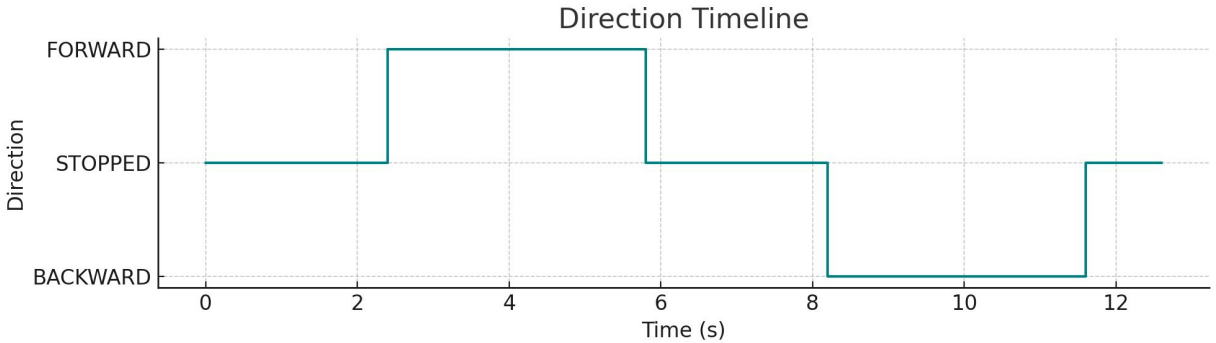
The HC-SR04 Ultrasonic sensor was tested with a logic level converter to the Raspberry Pi.

The sensor produced accurate distance readings as I moved closer and farther away, confirming stable performance and reliable data for obstacle detection.

Sensor Feedback

Arian Golkarfard

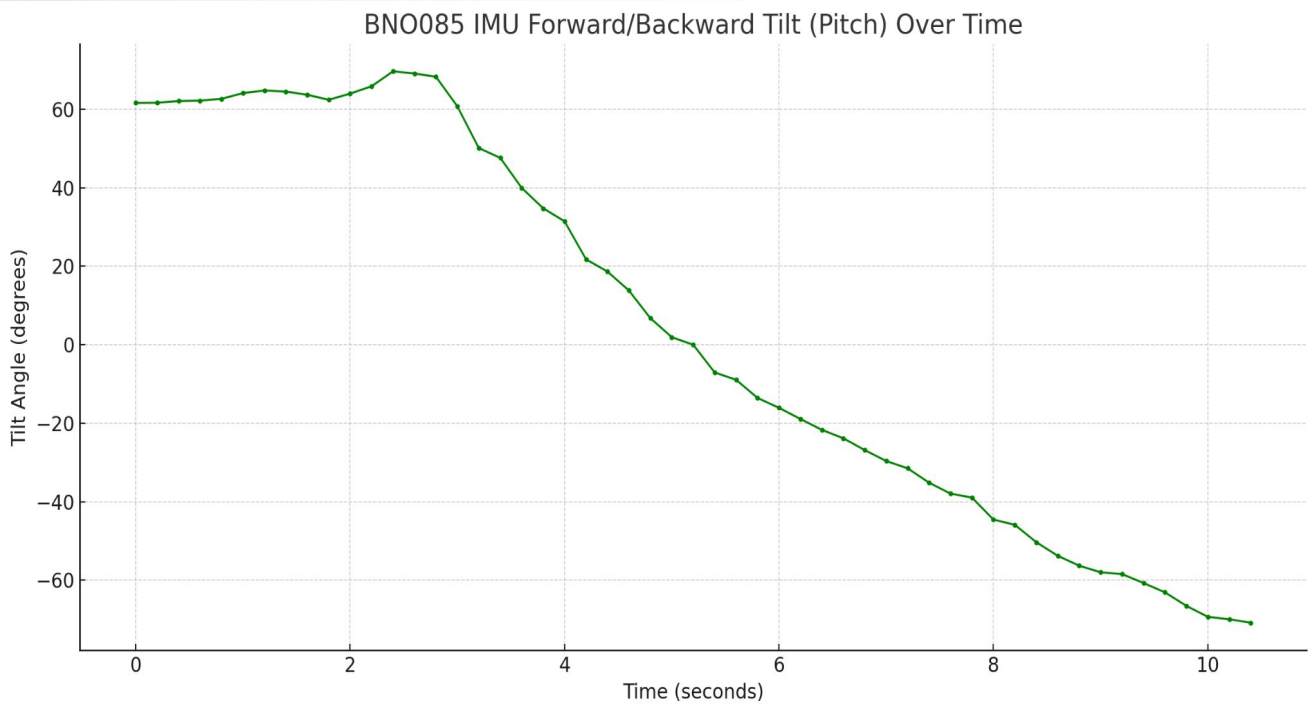


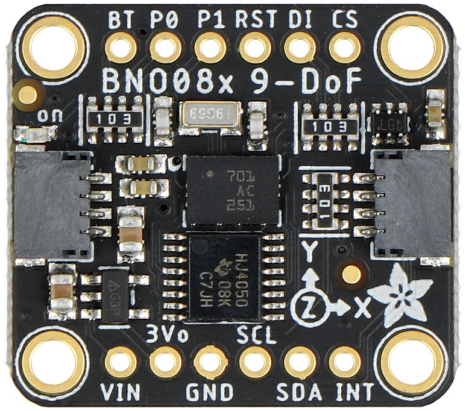


The AMT103 Encoder was tested by manually rotating it forward for about five seconds, stopping for two seconds, and then rotating it backward for another five seconds while recording data every 200 ms.

Sensor Feedback

Arian Golkarfard





The BNO085 IMU accurately tracked forward and backward tilt over time. The pitch angle smoothly transitioned from about +60° to -70°, showing consistent response and reliable measurement of orientation changes.

Web APP

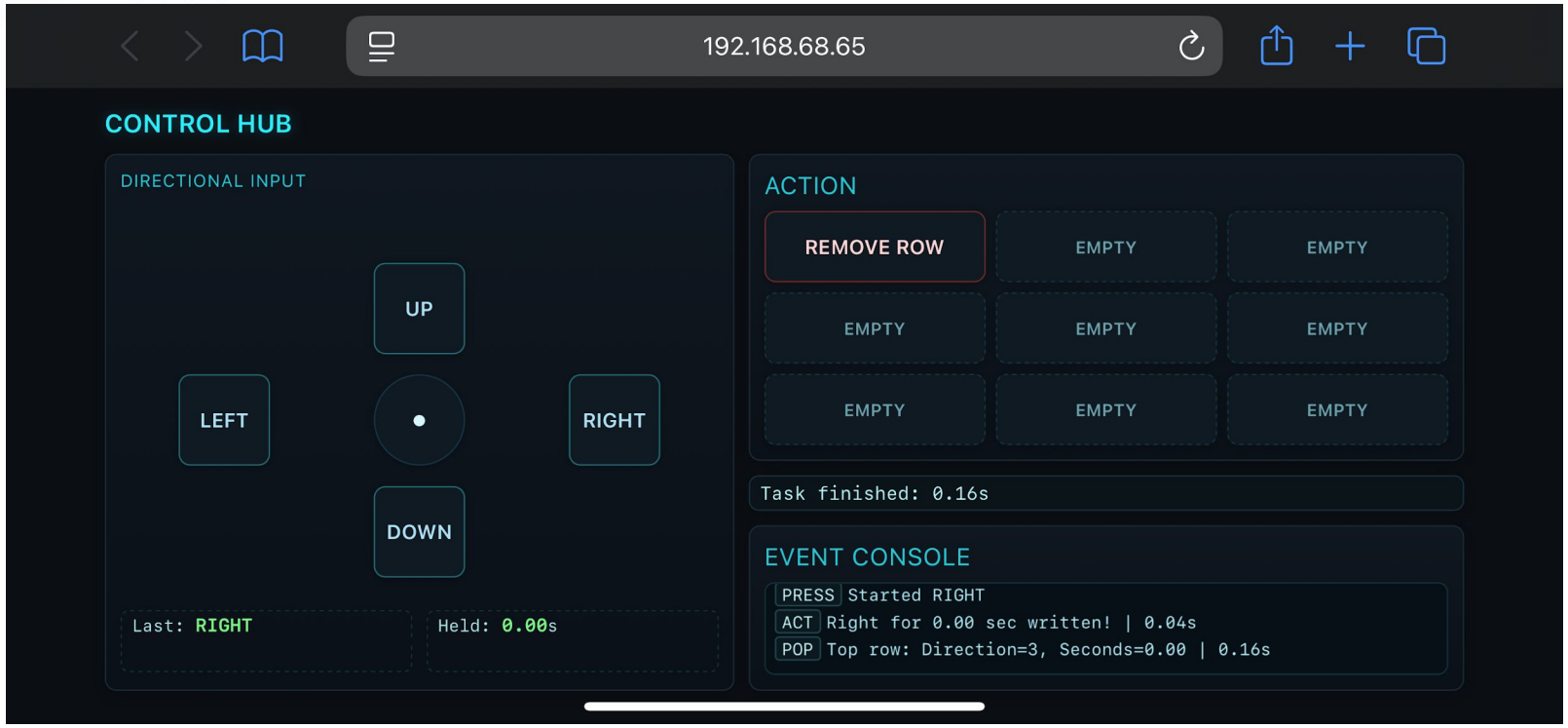
Sean Wall

Accomplishments since midterm update 30 hours	Ongoing progress/problems and plans until the next presentation
• Created a UI for the App • Created a Schematic and PCB for a connector board to allow easy sensor inputs • Assembled the bought bms to allow for battery charging	• Soldering the Connector Board • Creating connectors for all of the sensors • Start 3d modeling • Web app integration with other Programs

Web APP

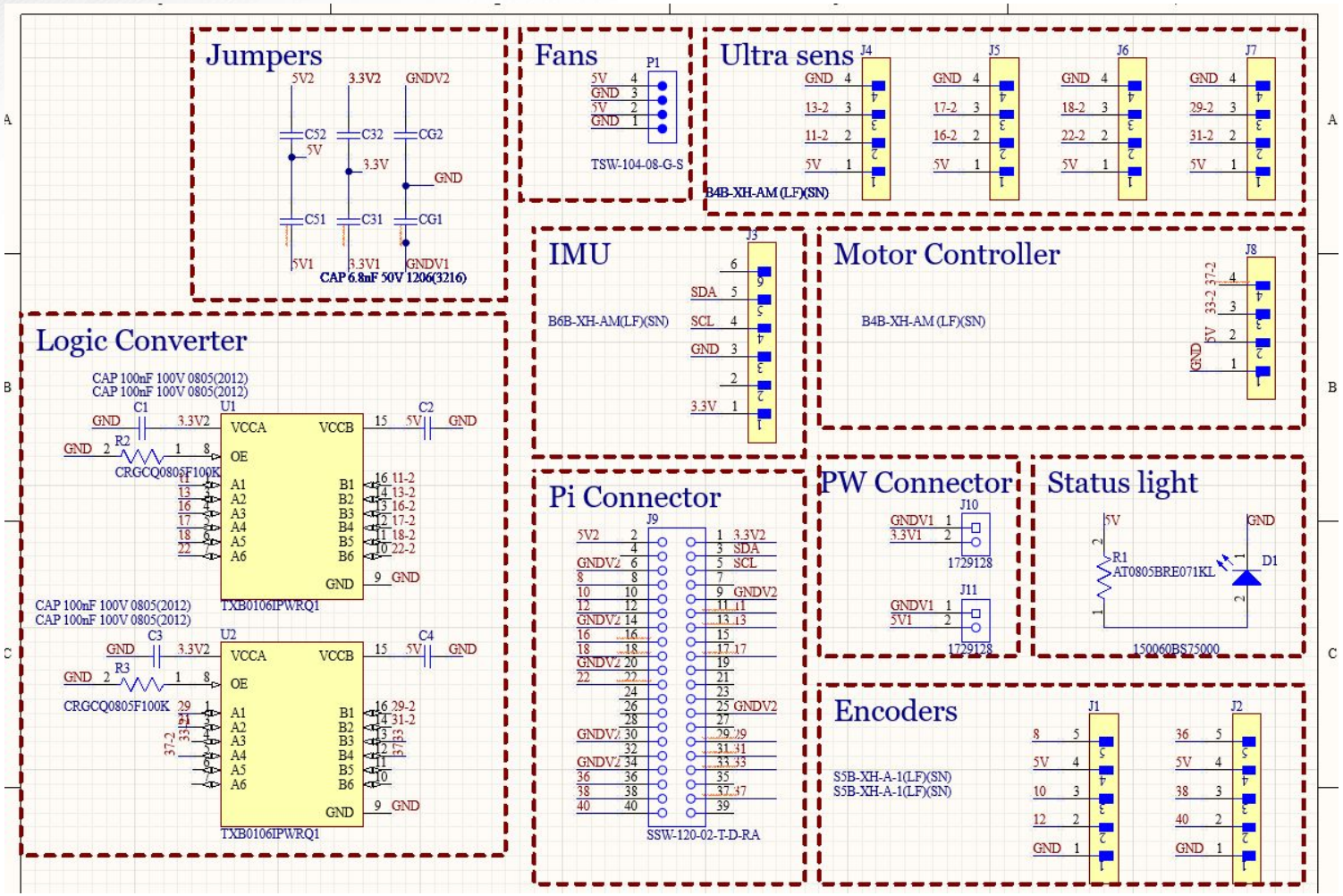
Sean Wall

- Can add data in around ~.05s
- Can Delete Data in around ~.15s
- Has the ability to add more functionality in the future
- It is already usable on my phone as seen here



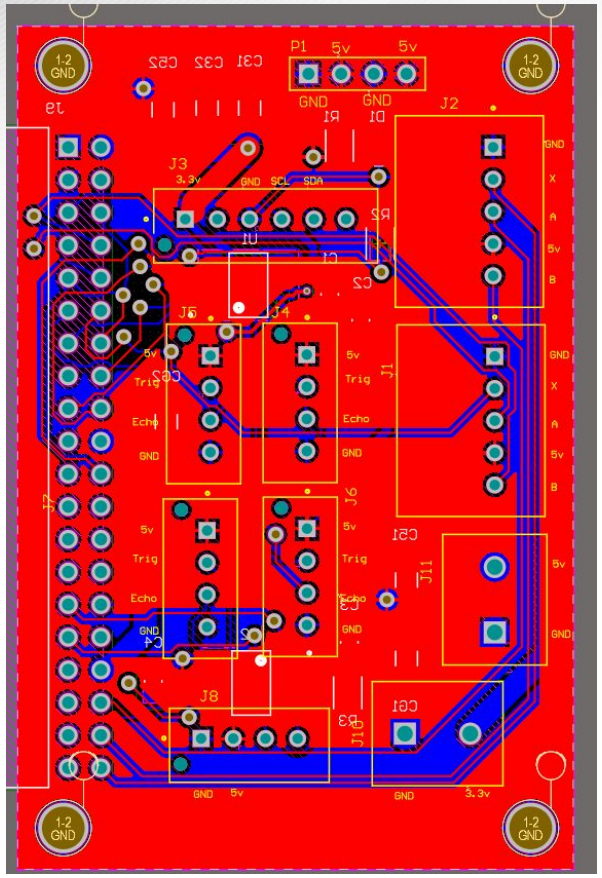
Connector Board Schematic

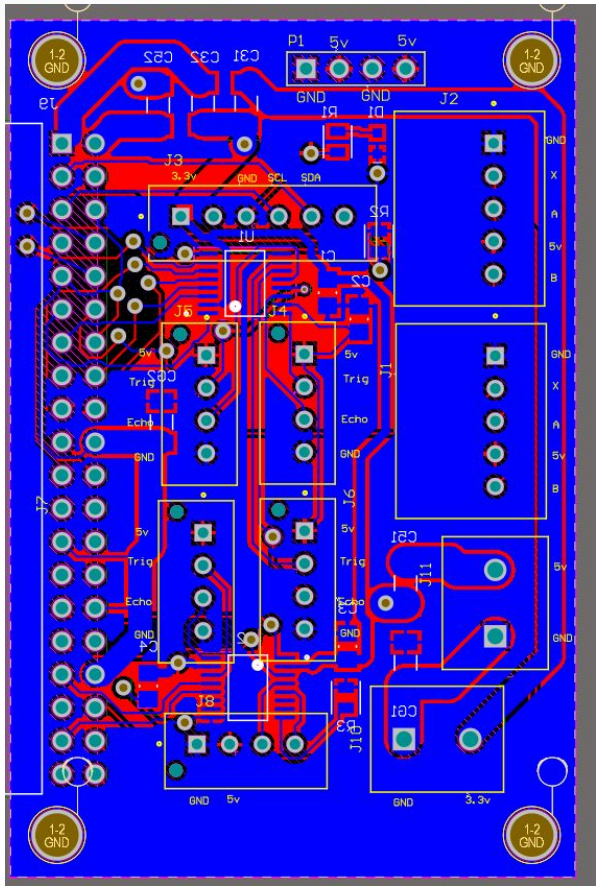
Sean Wall

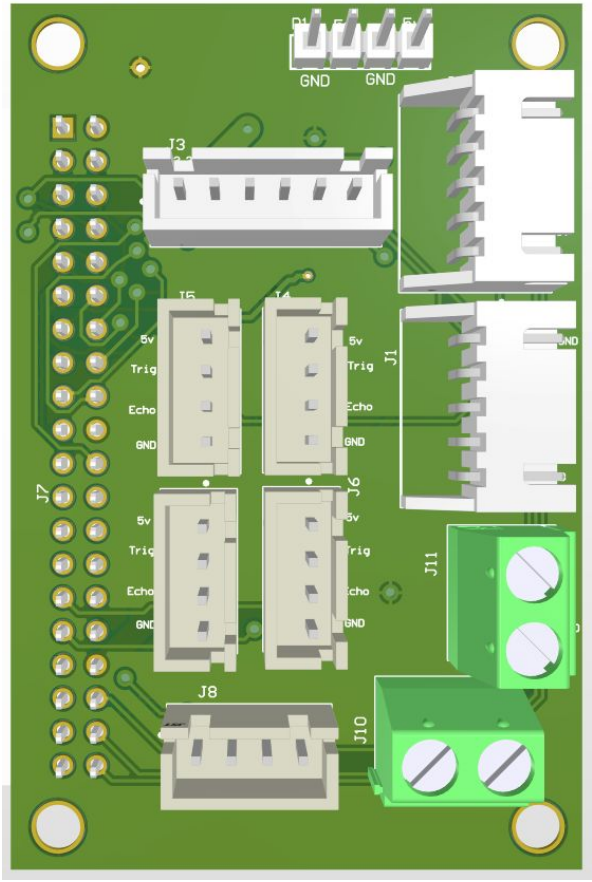


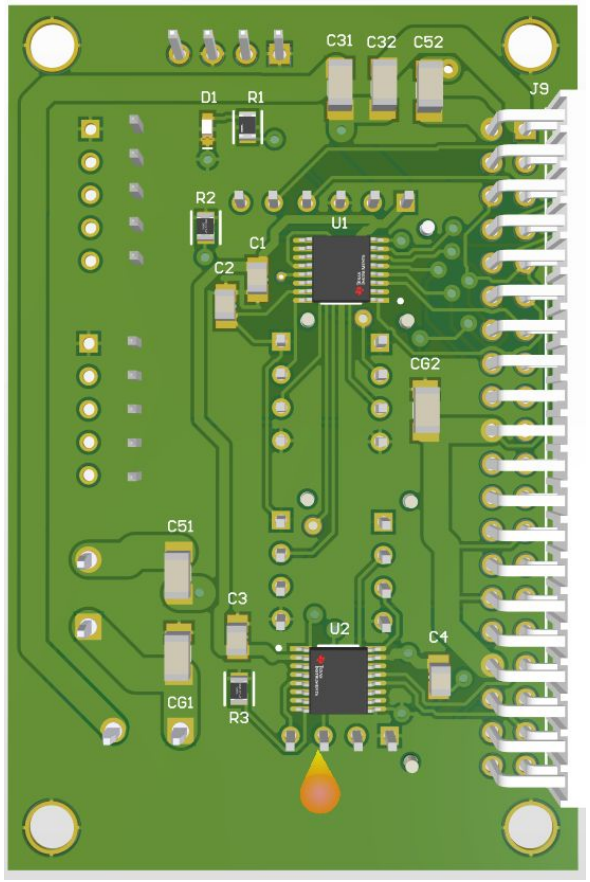
Connector Board PCB

Sean Wall





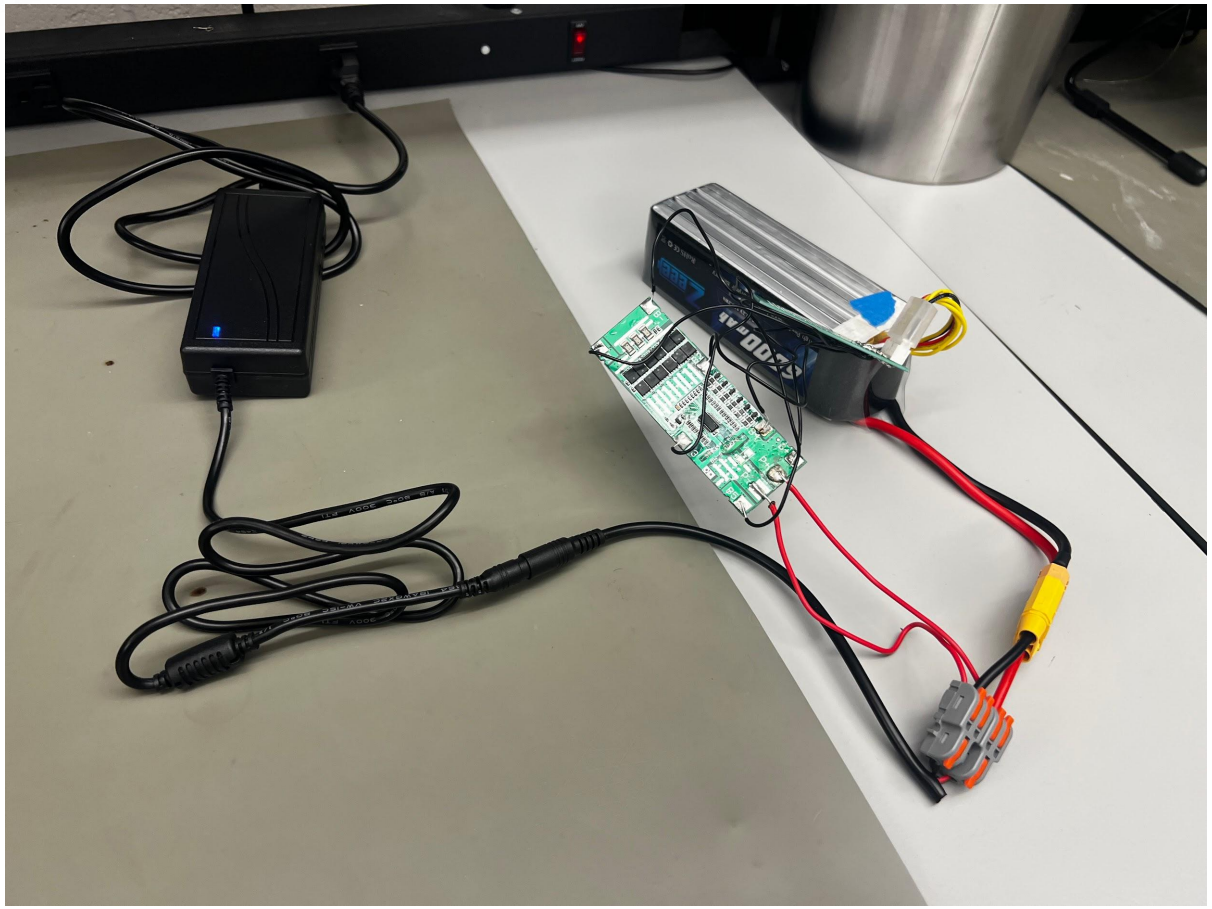




BMS Assembly

Sean Wall

- The battery has been successfully connected to the BMS and is able to be charged
- More data will be gathered in the future to determine how quickly the battery can be charged
- This data will be useful to know, but will not impact the success of the robot



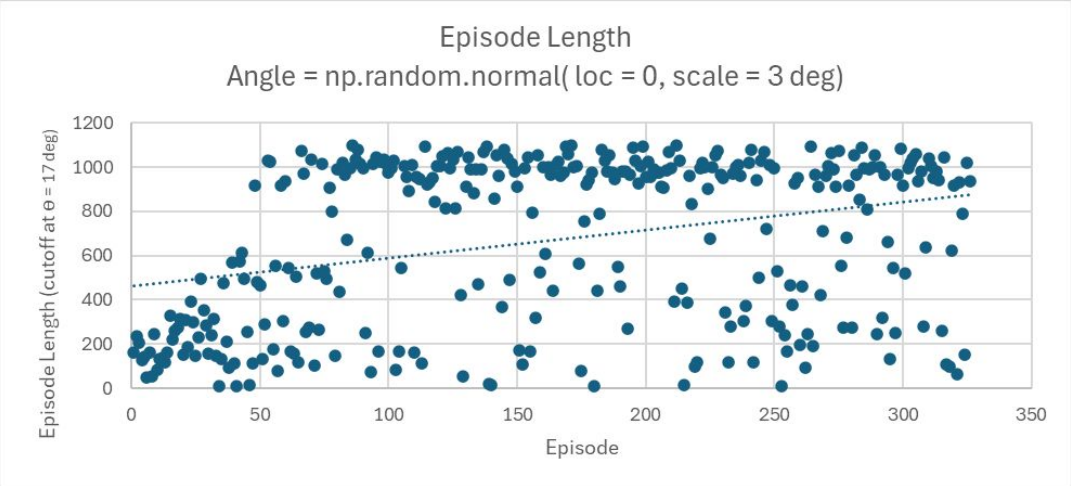
RL Agent

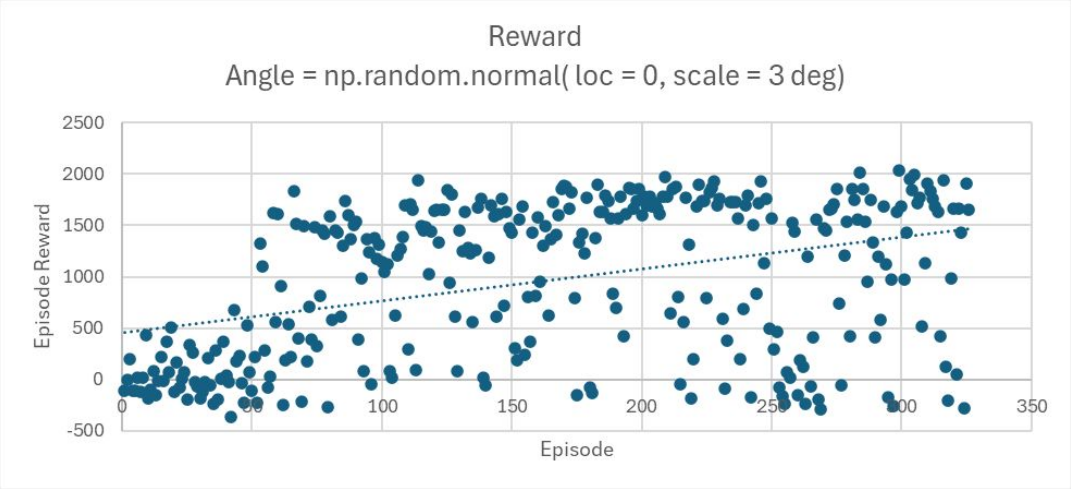
Dylan LeRoy

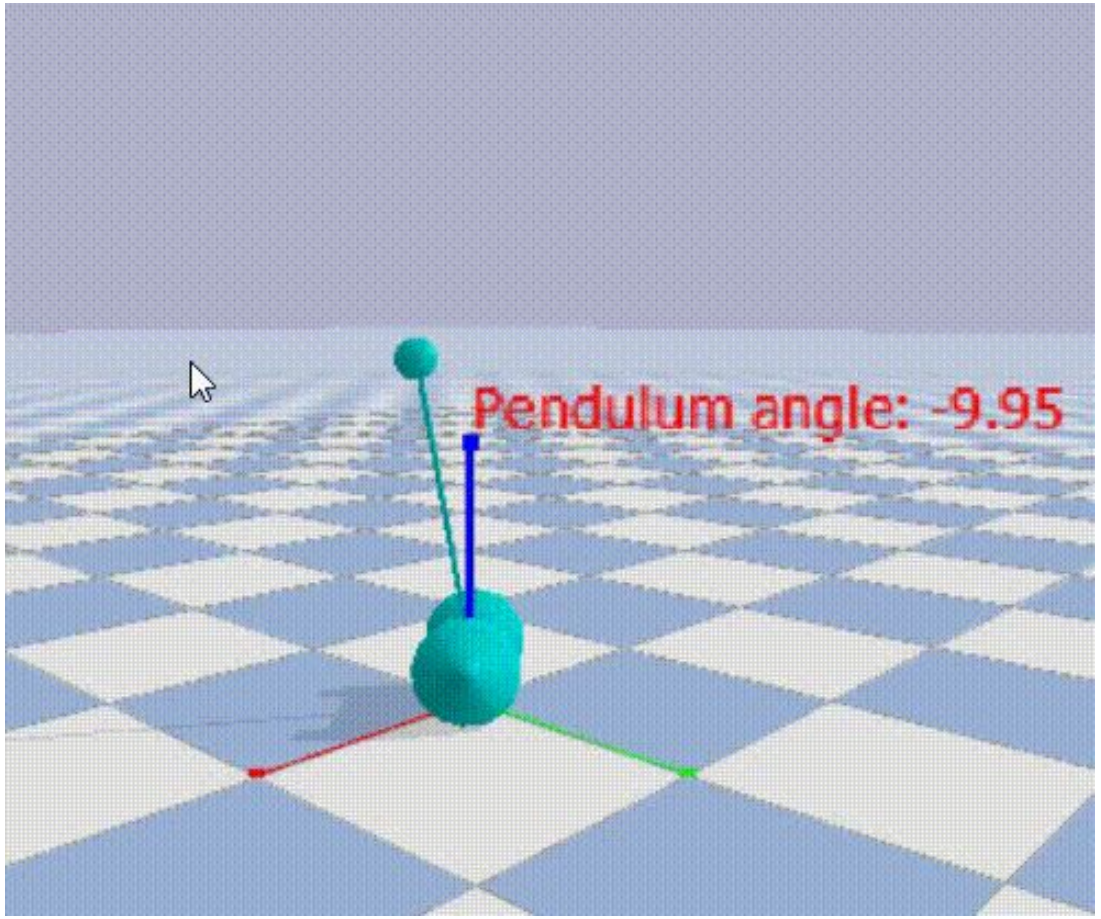
Accomplishments since midterm update 20-25 hours	Ongoing progress/problems and plans until the next presentation
• Create Gymnasium environment to let RL agent interact with robot in PyBullet • Create RL agent to begin training • Tweaks to reward function, normalize state inputs • Trained to Balance robot from starting upright +- 6 degrees • Weighed parts to realize more accurate model of our robot	• Train to correct from steeper angles • Train to correct from a push • Teach to drive and turn

RL Agent

Dylan LeRoy







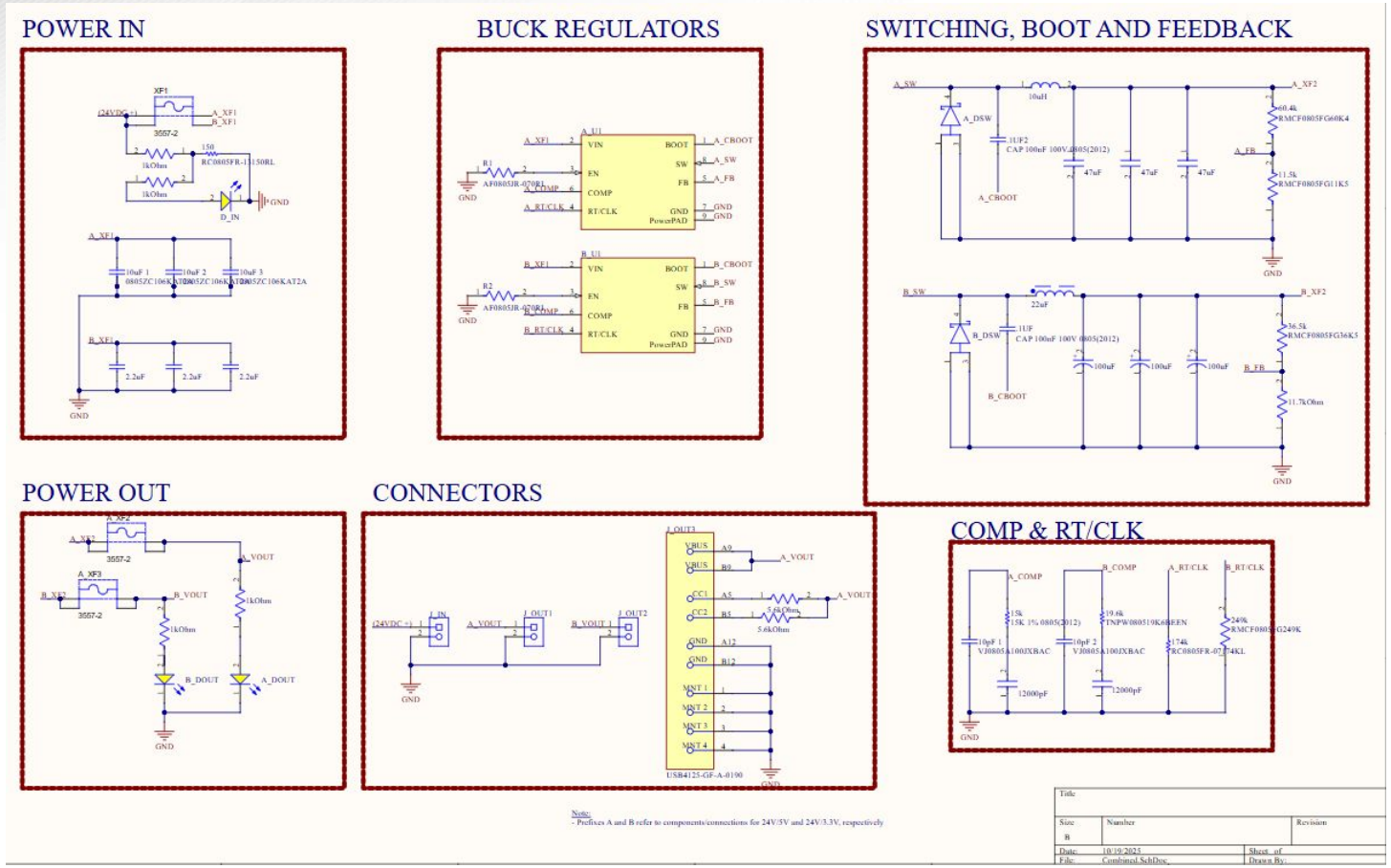
Power

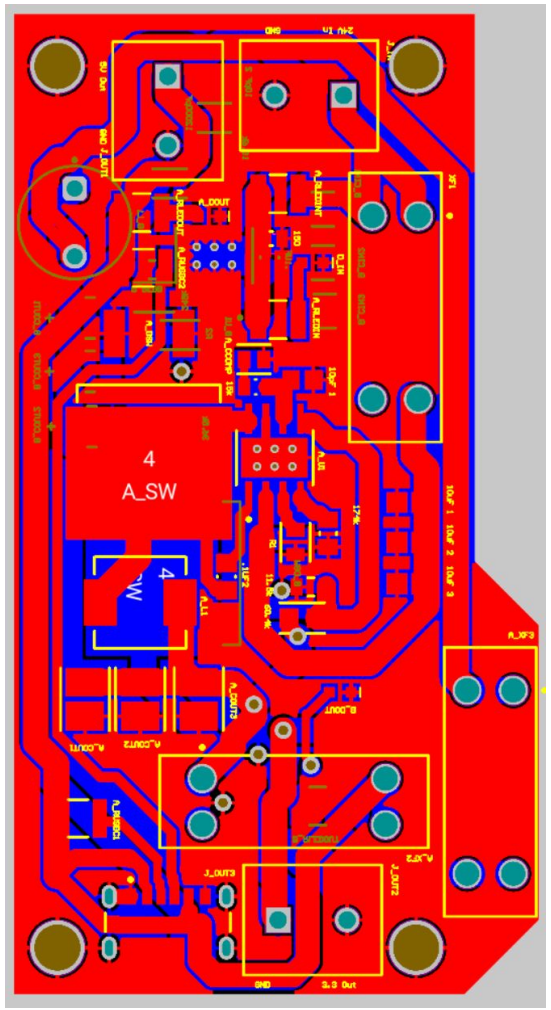
Stephen Darwin

Accomplishments since midterm update 20 Hours	Ongoing progress/problems and plans until the next presentation
• Verified buck converter functionality in LTSpice • Finalized buck converter component values • Finalized buck converter PCB layout in Altium • Ordered buck converter PCB and components	• Assembling and soldering buck converter PCB • Testing buck converter PCB to ensure accurate 5V/3.3V output voltages • Practicing with solder kits

Power

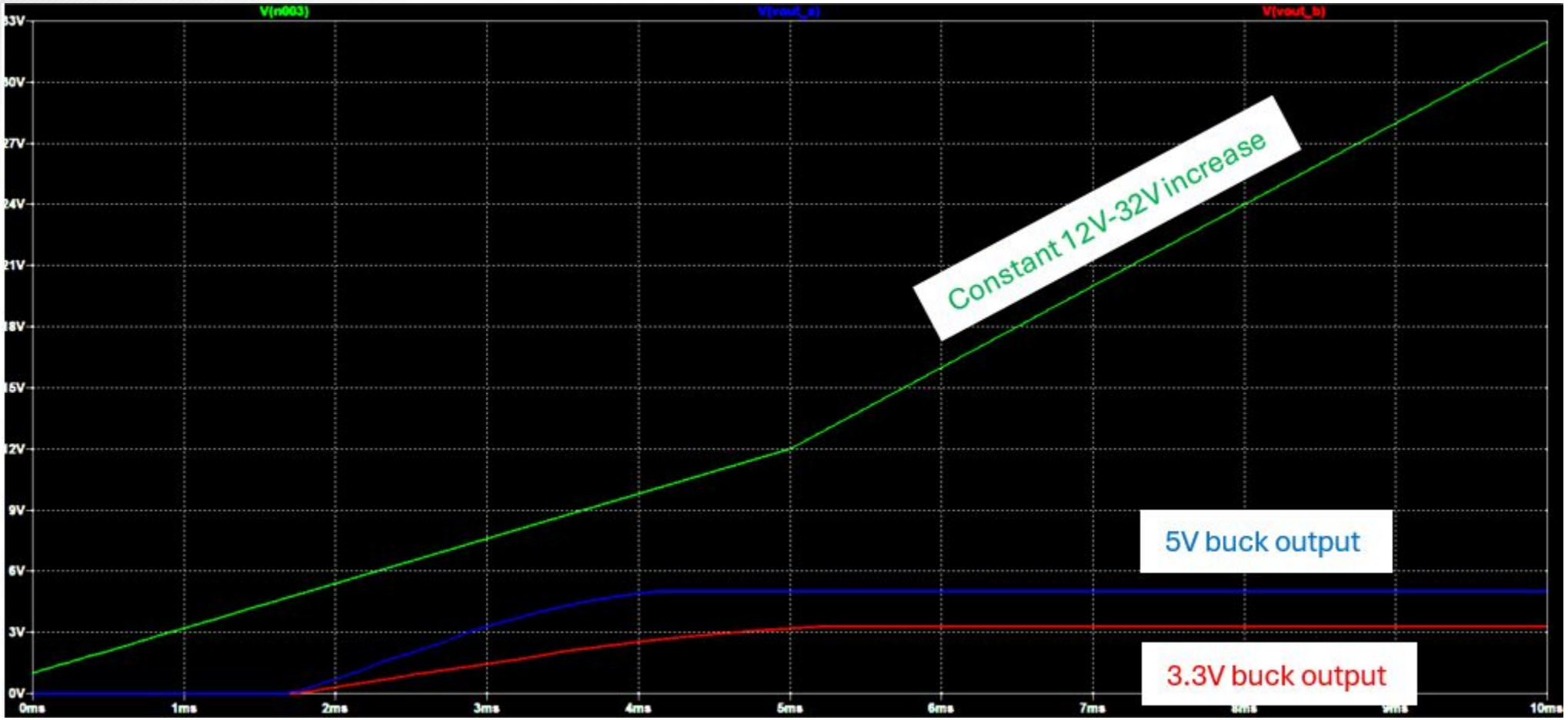
Stephen Darwin



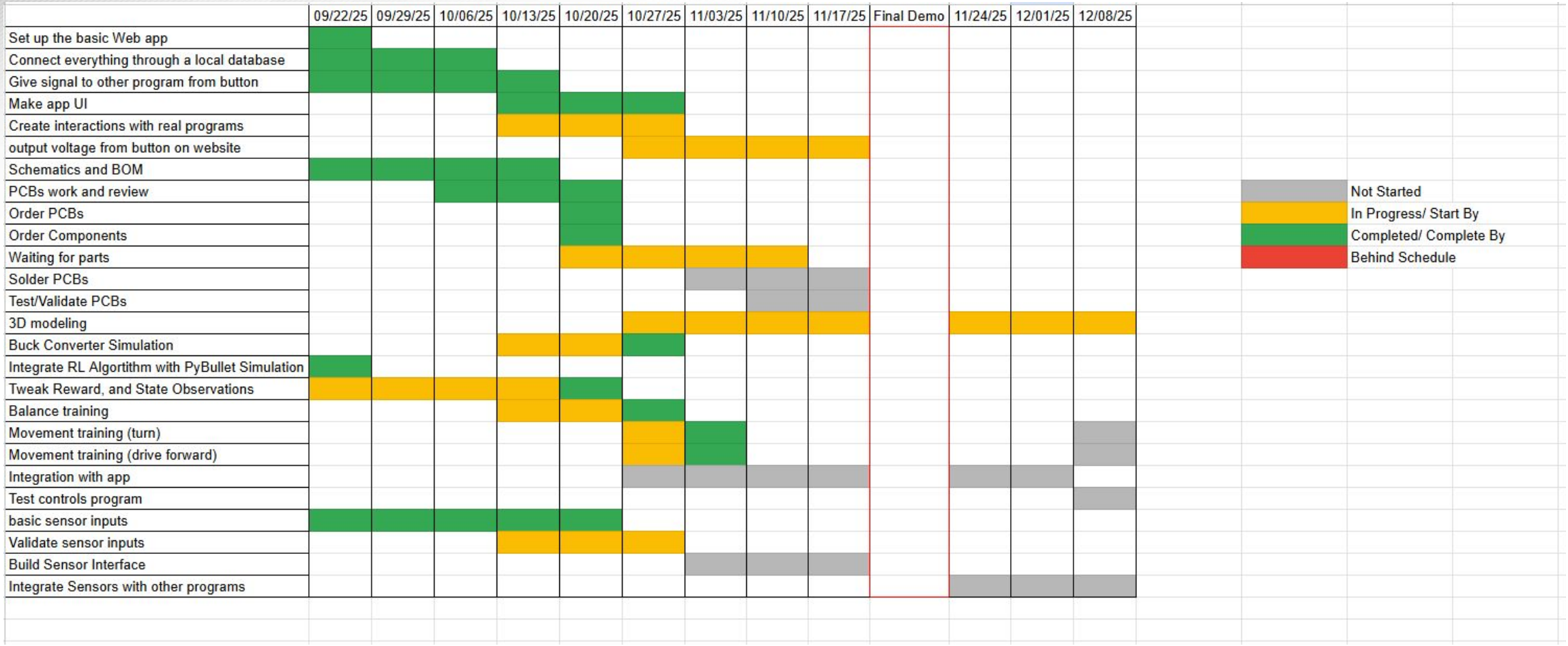


Power

Stephen Darwin



Execution plan



Validation plan

	10/29 - today	11/4	11/11	11/18	11/19
RL Controls System (Dylan)	>90% Pass Rate for correcting from angle of <10 degrees	Ensure Set Velocity maps directly to actual velocity	Ensure set turning rate maps to actual turning rate	Ensure proper interaction between other subsystems	
Web App/Database (Sean)	Start Integrating Web app with other programs	Solder connection board and test when it arrives	Make real programs controlled by App	Prepare a demo to show the app works with the gpio	Prepare Web app for final demo
Power (Stephen)	Waiting for buck PCB's to arrive	Solder buck PCBs	Test/validate buck PCBs to ensure accurate 5V and 3.3V outputs at variable input voltage	Prepare buck converters for final demo	Final demo
Sensor Feedback & Microcontroller (Arian)	Validated IMU sensor readings and confirmed stable pitch/tilt tracking.	Integrate all sensors and verify synchronized data output.	Calibrate sensors, lower polling rate to 30 ms, and test data stability.	Connect sensors to control boards and test full data pipeline.	Conduct final demo test with SAC reinforcement learning control system.

Thank You!